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United States Patent

12417978

Kind Code

B2

Date of Patent

September 16, 2025

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Microelectronic assemblies having backside die-to-package interconnects

Abstract

Microelectronic assemblies, related devices and methods, are disclosed herein. In some embodiments, a microelectronic assembly may include a first die, having a first surface and an opposing second surface, in a first layer, and including a first metallization stack at the first surface; a device layer on the first metallization stack; a second metallization stack on the device layer; and an interconnect on the first surface of the die electrically coupled to the first metallization stack; a conductive pillar in the first layer; and a second die, having a first surface and an opposing second surface, in a second layer on the first layer, wherein the first surface of the second die is coupled to the conductive pillar and to the second surface of the first die by a hybrid bonding region.

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Appl. No.: 17/470189

Filed: September 09, 2021

Prior Publication Data

Document Identifier

US 20230073026 A1

Publication Date

Mar. 09, 2023

Publication Classification

Int. Cl.: H01L23/528 (20060101); H01L23/522 (20060101)

U.S. Cl.:

CPC **H01L23/5283** (20130101); **H01L23/5226** (20130101);

Field of Classification Search

CPC: H01L (23/5283); H01L (23/5226)

USPC: 257/691

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Background/Summary

BACKGROUND

(1) For reliable operation of integrated circuit (IC) packages, as well as increased manufacturing assembly yields and reduced costs, IC dies and subassemblies may be tested prior to coupling to a package substrate or to each other so that only known good dies and subassemblies are used.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, not by way of limitation,

in the figures of the accompanying drawings.

(2) FIG. 1 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

(3) FIG. 2 is a side, cross-sectional exploded view of a portion of the microelectronic assembly of FIG. 1, in accordance with various embodiments.

(4) FIG. 3 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

(5) FIG. 4 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

(6) FIG. 5 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

(7) FIG. 6 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

(8) FIG. 7 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

(9) FIGS. 8A-8J are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 3, in accordance with various embodiments.

(10) FIGS. 9A-9D are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 1, in accordance with various embodiments.

(11) FIGS. 10A-10G are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 5, in accordance with various embodiments.

(12) FIGS. 11A-11D are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 7, in accordance with various embodiments.

(13) FIG. 12 is a top view of a wafer and dies that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

(14) FIG. 13 is a cross-sectional side view of an IC device that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

(15) FIG. 14 is a cross-sectional side view of an IC device assembly that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

(16) FIG. 15 is a block diagram of an example electrical device that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

(17) Microelectronic assemblies, related devices and methods, are disclosed herein. For example, in some embodiments, a microelectronic assembly may include a first die, having a first surface and an opposing second surface, in a first layer, and including a first metallization stack at the first surface; a device layer on the first metallization stack; a second metallization stack on the device layer; and a die-to-package interconnect on the first surface of the die electrically coupled to the first metallization stack; a conductive pillar in the first layer; and a second die, having a first surface and an opposing second surface, in a second layer on the first layer, wherein the first surface of the second die is coupled to the conductive pillar and to the second surface of the first die by a hybrid bonding region.

(18) Coupling two or more components by direct bonding in a multi-die IC package is challenging due to the increasingly small size and thickness of such components, the finer pitch of interconnects, and the reduced thickness of the bonding interface between components (e.g., z-height of die-to-die spacing), among others. Conventional methods for testing die functionality (e.g., to identify known good dies (KGD) during manufacturing) include using standard probing techniques to land on die pads. However, once dies are integrated into subassemblies, die pads may not be available for testing until after the subassembly is integrated into an IC package and thick metal layers (e.g., backside connections) are formed for connecting to a circuit board. In one example, a top wafer and a bottom wafer may be manufactured up to the fine pitch bonding layers,

the fine pitch bonding layers of the top and bottom wafers may be attached using wafer-to-wafer bonding techniques, then the backside of the bottom wafer may be thinned to reveal TSVs in the bottom die, and thick metal layers may be formed and electrically coupled to the TSVs, which enable testing for functionality. In another example, the fine pitch bonding layers of a top die and a bottom wafer may be attached using die-to-wafer bonding techniques, then the backside of the bottom wafer may be thinned to reveal TSVs in the bottom die, and thick metal layers may be formed and electrically coupled to the TSVs, which enable testing for functionality. In many such cases, when a bad or non-functioning subassembly is attached to a package, the manufacturing defect units and costs are compounded. Various ones of the microelectronic assemblies disclosed herein may exhibit better assembly yields during manufacturing, and improved performance and reliability during use, relative to conventional approaches by providing integrated interconnects to a backside of a die that may be used for testing die functionality in subassemblies. For example, the microelectronic assemblies disclosed herein may enable matching high-performance base dies to high performance top dies to achieve the best possible performance, customizing the backside power grid for each base die to reduce the cost or improve the performance, and separating the requirements on the base die signaling vias (e.g., backside-to-frontside vias) from the base die power delivery vias to optimize performance.

(19) In the following detailed description, reference is made to the accompanying drawings that form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense.

(20) Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

(21) For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The drawings are not necessarily to scale. Although many of the drawings illustrate rectilinear structures with flat walls and right-angle corners, this is simply for ease of illustration, and actual devices made using these techniques will exhibit rounded corners, surface roughness, and other features.

(22) The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. As used herein, a “package” and an “IC package” are synonymous, as are a “die” and an “IC die.” The terms “top” and “bottom” may be used herein to explain various features of the drawings, but these terms are simply for ease of discussion, and do not imply a desired or required orientation. As used herein, the term “insulating” means “electrically insulating,” unless otherwise specified. Throughout the specification, and in the claims, the term “coupled” means a direct or indirect connection, such as a direct electrical, mechanical, or magnetic connection between the things that are connected or an indirect connection, through one or more passive or active intermediary devices. The meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.”

(23) When used to describe a range of dimensions, the phrase “between X and Y” represents a range that includes X and Y. For convenience, the phrase “FIG. 8” may be used to refer to the

collection of drawings of FIGS. 8A-8J, the phrase “FIG. 9” may be used to refer to the collection of drawings of FIGS. 9A-9D, etc. Although certain elements may be referred to in the singular herein, such elements may include multiple sub-elements. For example, “an insulating material” may include one or more insulating materials.

(24) FIG. 1 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments. The microelectronic assembly **100** may include a multi-layer die subassembly **104** having integrated backside die-to-package (DTP) interconnects **150**. As used herein, the term a “multi-layer die subassembly” **104** may refer to a composite die having two or more stacked dielectric layers with one or more dies in each layer, and conductive interconnects and/or conductive pathways connecting the one or more dies, including dies in non-adjacent layers. As used herein, the terms a “multi-layer die subassembly” and a “composite die” may be used interchangeably. As shown in FIG. 1, the multi-layer die subassembly **104** may include a first layer **104-1** having a die **114-1** and conductive pillars **152**, and a second layer **104-2** having a die **114-2** and a die **114-3**. The first layer **104-1** may include a first surface **170-1** and an opposing second surface **170-2**. In particular, the multi-layer die subassembly **104** may include a first die **114-1** in a first dielectric layer **104-1**, a second die **114-2** in a second dielectric layer **104-2** coupled to the first die **114-1** by a first hybrid bonding region **130-1**, and a third die **114-3** in the second dielectric layer **104-2** coupled to the first die **114-1** by a second hybrid bonding region **130-2**. The die **114-1** may include a first metallization stack **126** at a first surface **170-1**, a substrate layer **120** on the first metallization stack **126**, a device layer **124** having devices **125** on the substrate layer **120**, a second metallization stack **122** on the device layer (e.g., at a second surface **170-2**), and DTP interconnects **150** at the first surface **170-1** of the die **114-1** coupled to the first metallization stack **126**. The first and second metallization stacks **126**, **122** may include a plurality of layers that include an insulating material formed in multiple layers and multiple conductive pathways formed through the insulating material. The conductive pathways in the first and second metallization stacks **126**, **122** may include conductive traces and/or conductive vias. The first metallization stack **126** may be referred to herein as “backside metal layers,” “thick metallization layers,” or other like terms, and the second metallization stack **122** maybe be referred to herein as “active side metal layers” or “thin metallization layers,” or other like terms, where the conductive structures of the first metallization stack **126** may be thicker than the conductive structures of the second metallization stack **122**. The device layer **124** may include active and passive devices (e.g., transistors, diodes, resistors, inductors, and capacitors, among others). In some embodiments, the device layer **124** may include one or more device layers including transistors (e.g., as discussed below with reference to FIG. 13). For example, the device layer **124** may include first and second transistors, where the first transistor may be a p-type metal oxide semiconductor (PMOS) and the second transistor may be an n-type metal oxide semiconductor (NMOS). The substrate layer **120** may include micro-through silicon vias (μ TSVs) **123**. The μ TSVs **123** may connect the first metallization stack **126** to devices **125** in the device layer **124** through the second metallization stack **122**. In some embodiments, the μ TSVs **123** have a pitch between 0.01 microns and 0.5 microns. In some embodiments, the substrate layer **120** may be omitted.

(25) The die **114-1** may be coupled to the package substrate **102** by the backside DTP interconnects **150**. As used herein, the term “backside DTP interconnects” or “DTP interconnects” may include conductive contacts **132** on the multi-layer die subassembly **104** at the first surface **170-1** of the die **114-1** that are coupled to a first metallization stack **126** (e.g., backside metallization layers) of a die **114-1**, and may further include solder **134**, or other interconnect structures, and may further include conductive contacts **136** on a surface of a substrate (e.g., a silicon or glass interposer, a package substrate **102** or a circuit board (not shown) in the absence of a package substrate **102** between the multi-layer die subassembly **104** and the circuit board). As used herein, a “conductive contact” may refer to a portion of conductive material (e.g., metal) serving as an electrical interface between different components; conductive interconnects may be recessed in, flush with, or extending away

from a surface of a component, and may take any suitable form (e.g., a conductive pad or socket, or portion of a conductive line or via). The conductive contacts **132** on the surface of the die **114-1** may further be coupled to conductive pathways in the die **114-1** (e.g., coupled to devices **125** in the device layer **124** through the μ TSVs **123** in the substrate layer **120** and/or through the second metallization stack **122**). The DTP interconnects **150** may be configured to route power or signals to and from the dies **114** in the multi-layer die subassembly **104** through the conductive contacts **132**, the conductive pillars **152**, and/or the first metallization stack **126** in die **114-1**.

(26) The dies **114-2**, **114-3** in the second layer **104-2** may be coupled to the package substrate **102** through the conductive pillars **152** to form multi-level (ML) interconnects. In particular, the dies **114-2**, **114-3** may be coupled to the package substrate **102** through the conductive pillars **152**, the conductive contacts **132** on the multi-layer die subassembly **104** (e.g., at the first surface **170-1**), the solder **134**, and the conductive contacts **136** on the package substrate **102**. The ML interconnects may be power delivery interconnects or high speed signal interconnects. As used herein, the term “ML interconnect” may refer to an interconnect that includes a conductive pillar between a first component and a second component where the first component and the second component are not in adjacent layers, or may refer to an interconnect that spans one or more layers (e.g., an interconnect between a first die in a first layer and a second die in a third layer, or an interconnect between a package substrate and a die in a second layer). The dies **114** may include other conductive pathways (e.g., including lines and vias) and/or to other circuitry (not shown) coupled to the respective conductive contacts (e.g., conductive contacts **132** on die **114-1** and/or conductive contacts **110** on dies **114-1**, **114-2**, **114-3**).

(27) The microelectronic assembly **100** may include a second die **114-2** coupled to a first die **114-1** by a hybrid bonding (HB) region **130-1**. In particular, as illustrated in FIG. 2, the HB region **130-1** may include a HB interface **180-1A** at the top surface of the first die **114-1**, with the HB interface **180-1A** including a set of conductive HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-1A**. The HB region **130-1** may also include a HB interface **180-1B** at the bottom surface of the die **114-2**, with the HB interface **180-1B** including a set of HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-1B**. The HB contacts **110** of the HB interface **180-1A** of the die **114-1** may align with the HB contacts **110** of the HB interface **180-1B** of the die **114-2** so that, in the microelectronic assembly **100**, the HB contacts **110** of the die **114-2** are in contact with the HB contacts **110** of the die **114-1**. In the microelectronic assembly **100** of FIG. 1, the HB interface **180-1A** of the die **114-1** may be bonded (e.g., electrically and mechanically) with the HB interface **180-1B** of the die **114-2** to form the HB region **130-1** coupling the die **114-1** and the die **114-2**. The second die **114-2** may be further coupled to a conductive pillar **152** by the HB region **130-1**.

(28) The microelectronic assembly **100** may further include a third die **114-3** coupled to a first die **114-1** by a hybrid bonding (HB) region **130-2**. In particular, as illustrated in FIG. 2, the HB region **130-2** may include a HB interface **180-2A** at the top surface of the first die **114-1**, with the HB interface **180-2A** including a set of conductive HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-2A**. The HB region **130-2** may also include a HB interface **180-2B** at the bottom surface of the die **114-3**, with the HB interface **180-2B** including a set of HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-2B**. The HB contacts **110** of the HB interface **180-2A** of the die **114-1** may align with the HB contacts **110** of the HB interface **180-2B** of the die **114-3** so that, in the microelectronic assembly **100**, the HB contacts **110** of the die **114-3** are in contact with the HB contacts **110** of the die **114-1**. In the microelectronic assembly **100** of FIG. 1, the HB interface **180-2A** of the die **114-1** may be bonded (e.g., electrically and mechanically) with the HB interface **180-2B** of the die **114-3** to form the HB region **130-2** coupling the die **114-1** and the die **114-3**. More generally, the HB regions **130** disclosed herein may include two complementary HB interfaces **180** bonded together; for ease of illustration, many of the subsequent drawings may omit the identification of the HB interfaces **180**

to improve the clarity of the drawings. The third die **114-3** may be further coupled to a conductive pillar **152** by the HB region **130-2**. In some embodiments, the second die **114-2** and/or third die **114-3** may not be coupled to a conductive pillar by a HB region. In such instances, the second die **114-2** and/or third die **114-3** may be coupled to a conductive pillar by other interconnects, such as metal-to-metal.

(29) As used herein, the term “hybrid bonding” is used to include techniques in which the HB dielectric **108** of opposing HB interfaces **180** are brought into contact first, then subject to heat and sometimes compression, or techniques in which the HB contacts **110** and the HB dielectric **108** of opposing HB interfaces **180** are brought into contact substantially simultaneously, then subject to heat and compression. In such techniques, the HB contacts **110** and the HB dielectric **108** at one HB interface **180** are brought into contact with the HB contacts **110** and the HB dielectric **108** at another HB interface **180**, respectively, and elevated pressures and/or temperatures may be applied to cause the contacting HB contacts **110** and/or the contacting HB dielectrics **108** to bond. HB interconnects may be capable of reliably conducting a higher current than other types of interconnects; for example, some conventional solder interconnects may form large volumes of brittle IMCs when current flows, and the maximum current provided through such interconnects may be constrained to mitigate mechanical failure. Although FIGS. **1** and **2** show the HB dielectric **108** as extending fully along the entire top surface of the first dielectric layer **104-1**, the HB dielectric **108** may extend along only a portion of the bottom surfaces of the second and third dies **114-2**, **114-3** where the second and third dies **114-2**, **114-3** overlap the first die **114-1**.

(30) A HB dielectric **108** may include one or more dielectric materials, such as one or more inorganic dielectric materials. For example, a HB dielectric **108** may include silicon and nitrogen (e.g., in the form of silicon nitride); silicon and oxygen (e.g., in the form of silicon oxide); silicon, carbon, and nitrogen (e.g., in the form of silicon carbon nitride); silicon, carbon and oxygen (e.g., in the form of a carbon-doped silicon oxide); silicon, oxygen, and nitrogen (e.g., in the form of silicon oxynitride); aluminum and oxygen (e.g., in the form of aluminum oxide); titanium and oxygen (e.g., in the form of titanium oxide); hafnium and oxygen (e.g., in the form of hafnium oxide); silicon, oxygen, carbon, and hydrogen (e.g., in the form of tetraethyl orthosilicate (TEOS)); zirconium and oxygen (e.g., in the form of zirconium oxide); niobium and oxygen (e.g., in the form of niobium oxide); tantalum and oxygen (e.g., in the form of tantalum oxide); and combinations thereof.

(31) A HB contact **110** may include a pillar, a pad, or other structure. The HB contacts **110**, although depicted in the accompanying drawings in the same manner at both HB interfaces **180** of a HB region **130**, may have a same structure at both HB interfaces **180**, or the HB contacts **110** at different HB interfaces **180** may have different structures. For example, in some embodiments, a HB contact **110** in one HB interface **180** may include a metal pillar (e.g., a copper pillar), and a complementary HB contact **110** in a complementary HB interface **180** may include a metal pad (e.g., a copper pad) recessed in a dielectric. The HB pads also may have different shapes (e.g., a larger regular polygon on a HB interface and a small regular polygon on a complementary HB interface). A HB contact **110** may include any one or more conductive materials, such as copper, manganese, titanium, gold, silver, palladium, nickel, copper and aluminum (e.g., in the form of a copper aluminum alloy), tantalum (e.g., tantalum metal, or tantalum and nitrogen in the form of tantalum nitride), cobalt, cobalt and iron (e.g., in the form of a cobalt iron alloy), or any alloys of any of the foregoing (e.g., copper, manganese, and nickel in the form of manganin). The pad structure also may include a plurality of metals (e.g., may include a high conductivity metal, such as copper or aluminum, capped with a corrosion resistant metal, such as titanium or gold, or a corrosion resistant alloy, such as manganin). In some embodiments, the HB dielectric **108** and the HB contacts **110** of a HB interface **180** may be manufactured using low-temperature deposition techniques (e.g., techniques in which deposition occurs at temperatures below 250 degrees Celsius, or below 200 degrees Celsius), such as low-temperature plasma-enhanced chemical vapor

deposition (PECVD).

(32) FIG. 1 also illustrates the die **114-1** coupled to the package substrate **102** by backside DTP interconnects **150**. Although FIG. 1 depicts a particular number of dies **114** coupled to the package substrate **102** and to other dies **114** by HB regions **130**, this number and arrangement are simply illustrative, and a microelectronic assembly **100** may include any desired number and arrangement of dies **114** coupled to a package substrate **102** and to other dies **114** by HB regions **130**. Although a single reference numeral “**108**” is used to refer to the HB dielectrics of multiple different HB interfaces **180** (and different HB regions **130**), this is simply for ease of illustration, and the HB dielectric **108** of different HB interfaces **180** (even within a single HB region **130**) may have different materials and/or structures. Similarly, although a single reference numeral “**110**” is used to refer to the HB contacts of multiple different HB interfaces **180** (and different HB regions **130**), this is simply for ease of illustration, and the HB contacts **110** of different HB interfaces **180** (even within a single HB region **130**) may have different materials and/or structures.

(33) The die **114** disclosed herein may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and multiple conductive pathways formed through the insulating material. In some embodiments, the insulating material of a die **114** may include a dielectric material, such as silicon dioxide, silicon nitride, oxynitride, polyimide materials, glass reinforced epoxy matrix materials, or a low-k or ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, organic polymeric dielectrics, photo-imagable dielectrics, and/or benzocyclobutene-based polymers). In some embodiments, the insulating material of a die **114** may include a semiconductor material, such as silicon, germanium, or a III-V material (e.g., gallium nitride), and one or more additional materials. For example, an insulating material may include silicon oxide or silicon nitride. The conductive pathways in a die **114** may include conductive traces and/or conductive vias, and may connect any of the conductive contacts in the die **114** in any suitable manner (e.g., connecting multiple conductive contacts on a same surface or on different surfaces of the die **114**). Example structures that may be included in the dies **114** disclosed herein are discussed below with reference to FIG. 13. The conductive pathways in the dies **114** may be bordered by liner materials, such as adhesion liners and/or barrier liners, as suitable. In some embodiments, the die **114** is a wafer. In some embodiments, the die **114** is a monolithic silicon, a fan-out or fan-in package die, or a die stack (e.g., wafer stacked, die stacked, or multi-layer die stacked).

(34) In some embodiments, the die **114** may include conductive pathways to route power, ground, and/or signals to/from other dies **114** included in the microelectronic assembly **100**. For example, the die **114-1** may include TSVs, including a conductive material via, such as a metal via, isolated from the surrounding silicon or other semiconductor material by a barrier oxide, or other conductive pathways through which power, ground, and/or signals may be transmitted between the package substrate **102** and one or more dies **114** “on top” of the die **114-1** (e.g., in the embodiment of FIG. 1, the dies **114-2** and/or **114-3**). In some embodiments, the die **114-1** may not route power and/or ground to the dies **114-2** and **114-3**; instead, the dies **114-2**, **114-3** may couple directly to power and/or ground lines in the package substrate **102** by ML interconnects (e.g., through conductive contacts **132** and conductive pillars **152**). In some embodiments, the die **114-1** in the first layer **104-1**, also referred to herein as “base die,” “interposer die,” or bridge die,” may be thicker than the dies **114-2**, **114-3** in the second layer **104-2**. In some embodiments, a die **114** may span multiple layers of the multi-layer die subassembly **104** (e.g., may span first and second layers **104-1**, **104-2**). The die **114-1** of the microelectronic assembly **100** may be a single-sided die (in the sense that the die **114-1** only has conductive contacts on a single surface), or, as shown, may be a double-sided die (in the sense that the die **114-1** has conductive contacts on two surfaces (e.g., a top surface and a bottom surface)), and may be a mixed-pitch die (in the sense that the die **114-1** has sets of conductive contacts with different pitches). In some embodiments, the dies **114-2** and/or **114-3** may not include active devices or routing, and may simply provide thermal and/or

mechanical support. In such an embodiment, the HB regions **130-1** and/or **130-2** may not include HB contacts **110**. In some embodiments, the dies **114-2**, **114-3** may include the elements of the die **114-1** (e.g., a first metallization stack **126**, a device layer **124** having devices **125**, and a second metallization stack **122**). In some embodiments, the die **114-1** may be a memory device (e.g., as described below with reference to the die **1502** of FIG. **12**), a high frequency serializer and deserializer (SerDes), such as a Peripheral Component Interconnect (PCI) express. In some embodiments, the die **114-1** may be a processing die, a radio frequency chip, a power converter, a network processor, a workload accelerator, or a security encryptor. In some embodiments, the die **114-2** and/or the die **114-3** may be a processing die.

(35) The multi-layer die subassembly **104** may include an insulating material **133** (e.g., a dielectric material formed in multiple layers, as known in the art) to form the multiple layers and to embed one or more dies in a layer. In some embodiments, the insulating material **133** of the multi-layer die subassembly **104** may be a dielectric material, such as an organic dielectric material, a fire retardant grade 4 material (FR-4), bismaleimide triazine (BT) resin, polyimide materials, glass reinforced epoxy matrix materials, or low-k and ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). In some embodiments, the die **114** may be embedded in an inhomogeneous dielectric, such as stacked dielectric layers (e.g., alternating layers of different inorganic dielectrics). In some embodiments, the insulating material **133** of the multi-layer die subassembly **104** may be a mold material, such as an organic polymer with inorganic silica particles. The multi-layer die subassembly **104** may include one or more ML interconnects through the dielectric material (e.g., including conductive vias and/or conductive pillars, as shown). The multi-layer die subassembly **104** may have any suitable dimensions. For example, in some embodiments, a thickness of the multi-layer die subassembly **104** may be between 100 um and 2000 um. In some embodiments, the multi-layer die subassembly **104** may be a composite die, such as stacked dies. The multi-layer die subassembly **104** may have any suitable number of layers, any suitable number of dies, and any suitable die arrangement. For example, in some embodiments, the multi-layer die subassembly **104** may have between 3 and 20 layers of dies. In some embodiments, the multi-layer die subassembly **104** may include a layer having between 2 and 50 dies.

(36) The package substrate **102** may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and one or more conductive pathways to route power, ground, and signals through the dielectric material (e.g., including conductive traces and/or conductive vias, as shown). In some embodiments, the insulating material of the package substrate **102** may be a dielectric material, such as an organic dielectric material, a fire retardant grade 4 material (FR-4), BT resin, polyimide materials, glass reinforced epoxy matrix materials, organic dielectrics with inorganic fillers or low-k and ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). In particular, when the package substrate **102** is formed using standard printed circuit board (PCB) processes, the package substrate **102** may include FR-4, and the conductive pathways in the package substrate **102** may be formed by patterned sheets of copper separated by build-up layers of the FR-4. The conductive pathways in the package substrate **102** may be bordered by liner materials, such as adhesion liners and/or barrier liners, as suitable. In some embodiments, the package substrate **102** may be formed using a lithographically defined via packaging process. In some embodiments, the package substrate **102** may be manufactured using standard organic package manufacturing processes, and thus the package substrate **102** may take the form of an organic package. In some embodiments, the package substrate **102** may be a set of redistribution layers formed on a panel carrier by laminating or spinning on a dielectric material, and creating conductive vias and lines by laser drilling and plating. In some embodiments, the package substrate **102** may be formed on a removable carrier using any suitable technique, such as a redistribution layer technique. Any method known in the art for fabrication of the package substrate **102** may be used, and for the sake

of brevity, such methods will not be discussed in further detail herein.

(37) In some embodiments, the package substrate **102** may be a lower density medium and the die **114** may be a higher density medium or have an area with a higher density medium. As used herein, the term “lower density” and “higher density” are relative terms indicating that the conductive pathways (e.g., including conductive interconnects, conductive lines, and conductive vias) in a lower density medium are larger and/or have a greater pitch than the conductive pathways in a higher density medium. In some embodiments, a higher density medium may be manufactured using a modified semi-additive process or a semi-additive build-up process with advanced lithography (with small vertical interconnect features formed by advanced laser or lithography processes), while a lower density medium may be a PCB manufactured using a standard PCB process (e.g., a standard subtractive process using etch chemistry to remove areas of unwanted copper, and with coarse vertical interconnect features formed by a standard laser process). In other embodiments, the higher density medium may be manufactured using semiconductor fabrication process, such as a single damascene process or a dual damascene process. In some embodiments, additional dies may be disposed on the top surface of the dies **114-2**, **114-3**. In some embodiments, additional components may be disposed on the top surface of the dies **114-2**, **114-3**. Additional passive components, such as surface-mount resistors, capacitors, and/or inductors, may be disposed on the top surface or the bottom surface of the package substrate **102**, or embedded in the package substrate **102**.

(38) The backside DTP interconnects **150** disclosed herein may take any suitable form. In some embodiments, the backside DTP interconnects **150** may include solder **134** (e.g., solder bumps or balls that are subject to a thermal reflow to form the interconnects), as shown. In some embodiments, the backside DTP interconnects **150** may include an anisotropic conductive material, such as an anisotropic conductive film or an anisotropic conductive paste. An anisotropic conductive material may include conductive materials dispersed in a non-conductive material. The DTP interconnects **150** may be direct metal-to-metal bonding, such as copper-to-copper bonding. In some embodiments, for example, when the package is a silicon interposer, the DTP interconnects **150** may include hybrid bonding.

(39) The microelectronic assembly **100** of FIG. **1** may also include an underfill material **127**. In some embodiments, the underfill material **127** may extend between the die **114-1** and the package substrate **102** around the associated backside DTP interconnects **150**. The underfill material **127** may be an insulating material, such as an appropriate epoxy material. In some embodiments, the underfill material **127** may include a capillary underfill, non-conductive film (NCF), or molded underfill. In some embodiments, the underfill material **127** may include an epoxy flux that assists with soldering the die **114-1** to the package substrate **102** when forming the backside DTP interconnects **150**, and then polymerizes and encapsulates the backside DTP interconnects **150**. The underfill material **127** may be selected to have a coefficient of thermal expansion (CTE) that may mitigate or minimize the stress between the multi-layer die subassembly **104** and the package substrate **102** arising from uneven thermal expansion in the microelectronic assembly **100**. In some embodiments, the CTE of the underfill material **127** may have a value that is intermediate to the CTE of the package substrate **102** (e.g., the CTE of the dielectric material of the package substrate **102**) and a CTE of the multi-layer die subassembly **104**.

(40) The microelectronic assembly **100** of FIG. **1** may also include a circuit board (not shown). The package substrate **102** may be coupled to the circuit board by second-level interconnects at the bottom surface of the package substrate **102**. The second-level interconnects may be any suitable second-level interconnects, including solder balls for a ball grid array arrangement, pins in a pin grid array arrangement or lands in a land grid array arrangement. The circuit board may be a motherboard, for example, and may have other components attached to it. The circuit board may include conductive pathways and other conductive contacts for routing power, ground, and signals through the circuit board, as known in the art. In some embodiments, the second-level

interconnects may not couple the package substrate **102** to a circuit board, but may instead couple the package substrate **102** to another IC package, an interposer, or any other suitable component. In some embodiments, the multi-layer die subassembly **104** may not be coupled to a package substrate **102**, but may instead be coupled to a circuit board, such as a PCB.

(41) Many of the elements of the microelectronic assembly **100** of FIG. **1** are included in other ones of the accompanying drawings; the discussion of these elements is not repeated when discussing these drawings, and any of these elements may take any of the forms disclosed herein. Further, a number of elements are illustrated in FIG. **1** as included in the microelectronic assembly **100**, but a number of these elements may not be present in a microelectronic assembly **100**. For example, in various embodiments, the underfill material **127** and the package substrate **102** may not be included. In some embodiments, individual ones of the microelectronic assemblies **100** disclosed herein may serve as a system-in-package (SiP) in which multiple dies **114** having different functionality are included. In such embodiments, the microelectronic assembly **100** may be referred to as an SiP.

(42) FIG. **3** is a side, cross-sectional view of another example microelectronic assembly, in accordance with various embodiments. The microelectronic assembly **100** may include a multi-layer die subassembly **104** having integrated backside DTP interconnects **150**. As shown in FIG. **3**, the multi-layer die subassembly **104** may include a first layer **104-1** having a die **114-1** and conductive pillars **152**, and a second layer **104-2** having a die **114-2**. In particular, the multi-layer die subassembly **104** may include a first die **114-1** in a first dielectric layer **104-1** and a second die **114-2** in a second dielectric layer **104-2** coupled to the first die **114-1** by a first hybrid bonding region **130**. The die **114-1** may include a first metallization stack **126** at a first surface **170-1**, a substrate layer **120** on the first metallization stack **126**, a device layer **124** having devices **125** on the substrate layer **120**, a second metallization stack **122** on the device layer (e.g., at a second surface **170-2**), and DTP interconnects **150** at the first surface **170-1** of the die **114-1** coupled to the first metallization stack **126**. In some embodiments, the substrate layer **120** may be omitted. The die **114-1** may be coupled to the package substrate **102** through the backside DTP interconnects **150** and the die **114-2** in the second layer **104-2** may be coupled to the package substrate **102** by ML interconnects.

(43) FIG. **4** is a side, cross-sectional view of another example microelectronic assembly, in accordance with various embodiments. The microelectronic assembly **100** may include a multi-layer die subassembly **104** couple to a die **114-3** at a first surface **170-1** and having integrated backside DTP interconnects **150**. As shown in FIG. **4**, the multi-layer die subassembly **104** may include a first layer **104-1** having a die **114-1** and conductive pillars **152**, and a second layer **104-2** having a die **114-2**. In particular, the multi-layer die subassembly **104** may include a first die **114-1** in a first dielectric layer **104-1**, a second die **114-2** in a second dielectric layer **104-2** coupled to the first die **114-1** by a first hybrid bonding region **130-1**, and a third die **114-3** coupled to a first surface **170-1** of the first dielectric layer **104-1** by a second hybrid bonding region **130-2** with DTP interconnects **150** at a bottom surface of the die **114-3**. The die **114-1** may include a first metallization stack **126** at a first surface **170-1**, a substrate layer **120** on the first metallization stack **126**, a device layer **124** having devices **125** on the substrate layer **120**, a second metallization stack **122** on the device layer (e.g., at a second surface **170-2**). In some embodiments, the substrate layer **120** may be omitted. The die **114-3** may be a double-sided die and may include TSVs **121** and/or other conductive pathways (not shown) for coupling to the package substrate **102** and the multi-layer die subassembly **104**. The DTP interconnects **150** at the bottom surface of the die **114-3** may be coupled to the first metallization stack **126** in the die **114-1** by conductive pathways (e.g., TSVs **121**) in the die **114-3**. The die **114-1** may be coupled to the package substrate **102** through the die **114-3** and the backside DTP interconnects **150** and the die **114-2** in the second layer **104-2** may be coupled to the package substrate **102** through the conductive pillars **152** to form ML interconnects and conductive pathways in the die **114-3**. In some embodiments, the die **114-2** may be non-

functional and may provide mechanical and/or thermal support. In such an embodiment, the first hybrid bonding region **130-1** may not include HB contacts **110**. Further, the die **114-3** may be a passive die including pass-through and redistribution routing.

(44) FIG. 5 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments. The microelectronic assembly **100** may include a multi-layer die subassembly **104** having integrated backside DTP interconnects **150**. As shown in FIG. 5, the multi-layer die subassembly **104** may include a first layer **104-1** having a die **114-1** and conductive pillars **152**, and a second layer **104-2** having a die **114-2** and a die **114-3**. The first layer **104-1** may include a first surface **170-1** and an opposing second surface **170-2**. In particular, the multi-layer die subassembly **104** may include a first die **114-1** in a first dielectric layer **104-1**, a second die **114-2** in a second dielectric layer **104-2** coupled to the first die **114-1** by a first hybrid bonding region **130-1**, and a third die **114-3** in the second dielectric layer **104-2** coupled to the first die **114-1** by a second hybrid bonding region **130-2**. The die **114-1** may include a first substrate layer **128** at a first surface **170-1** having TSVs **118**, a first metallization stack **126** on the first substrate layer **128**, a second substrate layer **120** on the first metallization stack **126**, a device layer **124** having devices **125** on the substrate layer **120**, a second metallization stack **122** on the device layer (e.g., at a second surface **170-2**), and DTP interconnects **150** at the first surface **170-1** coupled to the first metallization stack **126** in the first die **114-1** through the TSVs **118** in the first substrate layer **128**. In some embodiments, the TSVs **118** in the first substrate layer **128** may have a pitch between 5 microns and 100 microns. In some embodiments, the second substrate layer **120** may be omitted. The die **114-1** may be coupled to the package substrate **102** through the backside DTP interconnects **150** and the dies **114-2**, **114-3** in the second layer **104-2** may be coupled to the package substrate **102** through the ML interconnects.

(45) FIG. 6 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments. The microelectronic assembly **100** may include a multi-layer die subassembly **104** having integrated backside DTP interconnects **150**. As shown in FIG. 6, the multi-layer die subassembly **104** may include a first layer **104-1** having a die **114-1** and conductive pillars **152**, and a second layer **104-2** having a die **114-2**. The first layer **104-1** may include a first surface **170-1** and an opposing second surface **170-2**. In particular, the multi-layer die subassembly **104** may include a first die **114-1** in a first dielectric layer **104-1** and a second die **114-2** in a second dielectric layer **104-2** coupled to the first die **114-1** by a hybrid bonding region **130**. The die **114-1** may include a first substrate layer **128** at a first surface **170-1** having TSVs **118**, a first metallization stack **126** on the first substrate layer **128**, a second substrate layer **120** on the first metallization stack **126**, a device layer **124** having devices **125** on the substrate layer **120**, a second metallization stack **122** on the device layer (e.g., at a second surface **170-2**), and DTP interconnects **150** at the first surface **170-1** coupled to the first metallization stack **126** in the first die **114-1** through the TSVs **118** in the first substrate layer **128**. In some embodiments, the second substrate layer **120** may be omitted. The die **114-1** may be coupled to the package substrate **102** through the backside DTP interconnects **150** and the die **114-2** in the second layer **104-2** may be coupled to the package substrate **102** through the ML interconnects.

(46) FIG. 7 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments. The microelectronic assembly **100** may include a multi-layer die subassembly **104** having integrated backside DTP interconnects **150**. As shown in FIG. 7, the multi-layer die subassembly **104** may include a redistribution layer (RDL) **148** having DTP interconnects **150** on a bottom surface, a first layer **104-1** on the top surface of the RDL **148**, and a second layer **104-2** on the first layer **104-1**. The first layer **104-1** may include a first surface **170-1** and an opposing second surface **170-2**. In particular, the multi-layer die subassembly **104** may include a first dielectric layer **104-1**, the RDL **148** coupled to the first surface **170-1** of the first layer **104-1**, and the second dielectric layer **104-2** coupled to the second surface **170-2** of the first dielectric layer **104-1**. The first dielectric layer **104-1** may include a first die **114-1**, a second die

114-2, and conductive pillars **152** embedded therein and the second dielectric layer **104-2** may include a third die **114-3** embedded therein coupled to the first die **114-1** and the second die **114-2** by a hybrid bonding region **130**. The dies **114-1**, **114-2** may include a first substrate layer **128** at a first surface **170-1** having TSVs **118**, a first metallization stack **126** on the first substrate layer **128**, a second substrate layer **120** on the first metallization stack **126**, a device layer **124** having devices **125** on the substrate layer **120**, a second metallization stack **122** on the device layer (e.g., at a second surface **170-2**). In some embodiments, the second substrate layer **120** may be omitted. The first metallization stacks **126** in the respective dies **114-1**, **114-2** may be coupled to the DTP interconnects **150** on the bottom surface of the RDL **148** through the TSVs **118** in the first substrate layer **128** and conductive pathways in the RDL **148**. The dies **114-1**, **114-2** may be coupled to the package substrate **102** through the backside DTP interconnects **150** and the die **114-3** in the second layer **104-2** may be coupled to the package substrate **102** by the DTP interconnects **150** through the conductive pillars **152** of the ML interconnects. Although FIG. 7 shows a particular number and arrangement of a microelectronic assembly **100** including a plurality of embedded first, second, and third dies **114**, and a single RDL **148**, a microelectronic assembly **100** may include any number and arrangement of dies **114** and RDLs **148**, including two or more RDLs **148** and including an RDL **148** at the second surface **170-2** of the first dielectric layer **104-1**.

(47) Any suitable techniques may be used to manufacture the microelectronic assemblies **100** disclosed herein. For example, FIGS. **8A-8J** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly **100** of FIG. **3**, in accordance with various embodiments. Although the operations discussed below with reference to FIGS. **8A-8J** (and others of the accompanying drawings representing manufacturing processes) are illustrated in a particular order, these operations may be performed in any suitable order.

(48) FIG. **8A** illustrates an assembly subsequent to placing a first die **114-1** on a first carrier **105-1** with an active surface (e.g., metallization stack **122**) facing towards the first carrier **105-1**. The first die **114-1** may include an active side metallization stack **122**, a device layer **124** having a device **125**, and a substrate **120** having a μ TSV **123** (e.g., at a backside surface opposing the active surface), where the substrate **120** includes a non-electrical material on and over the μ TSV **123**. The non-electrical material, which is an inactive portion of the die **114-1**, may include silicon, germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, gallium antimonide, further materials classified as group III-V, or an insulating material, such as silicon dioxide (glass), ceramic, or quartz, among other materials. A carrier **105** may include any suitable material, and in some embodiments, may include a semiconductor wafer (e.g., a silicon wafer) or glass (e.g., a glass panel). The first die **114-1** may be attached to the first carrier **105-1** using any suitable technique, including a temporary adhesive layer or a die attach film (DAF).

(49) FIG. **8B** illustrates an assembly subsequent to removing the non-electrical material from the top surface of the substrate **120** and revealing the top surface of the μ TSV **123**. The non-electrical material may be removed using any suitable technique, including, for example, grinding, etching, such as reactive ion etching (RIE) or chemical etching. In some embodiments, the top surface of the substrate **120** may be polished to reveal the top surface of the μ TSV **123**. In some embodiments, when the μ TSV **123** are not included in the assembly of FIG. **8A**, the μ TSV **123** may be formed in the substrate material **120** subsequent to thinning the non-electrical material at the top surface of the substrate **120**. In some embodiments, the first die **114-1** may be processed at the wafer level and subsequently singulated.

(50) FIG. **8C** illustrates an assembly subsequent to forming a backside metallization stack **126** on the top surface of the assembly of FIG. **8B**, electrically coupling the backside metallization stack **126** and the active side metallization stack **122** by the μ TSV **123** in the substrate **120**, and forming conductive pads **142** on the top surface of the backside metallization stack **126**. The die **114-1** may be functionally tested using the conductive pads **142** or the top metal layer in the stack **126** to

determine that the die **114-1** is a KGD before further processing is performed.

(51) FIG. **8D** illustrates an assembly subsequent to mounting a second carrier **105-2** to the top surface of the assembly of FIG. **8C**.

(52) FIG. **8E** illustrates an assembly subsequent to inverting the assembly of FIG. **8D** and removing the first carrier **105-1**.

(53) FIG. **8F** illustrates an assembly subsequent to forming an exposed HB interface **180** on a top surface of the assembly of FIG. **8E** (e.g., on the active side metallization stack **122**), where the HB interface **180** includes HB contacts **110** surrounded by HB dielectric **108**.

(54) FIG. **8G** illustrates an assembly subsequent to hybrid bonding a second die **114-2** to the top surface of the assembly of FIG. **8F**. In particular, HB interface **180** (not labeled) of the second die **114-2** may be brought into contact with the HB interface of the first die **114-1**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB region **130**.

(55) FIG. **8H** illustrates an assembly subsequent to inverting the assembly of FIG. **8G** and removing the second carrier **105-2**.

(56) FIG. **8I** illustrates an assembly subsequent to forming conductive pillars **152**, depositing an insulating material **133** on and around the first die **114-1** and the conductive pillars **152**, and forming conductive contacts **132** for DTP interconnects on the top surfaces of the first die **114-1** and the conductive pillars **152**. The conductive pillars **152** may be formed using any suitable technique, for example, a lithographic process or an additive process, such as cold spray or 3-dimensional printing. For example, the conductive pillars **152** may be formed by depositing, exposing, and developing a photoresist layer on the top surface of the die **114-2**. The photoresist layer may be patterned to form cavities in the shape of the conductive pillars. Conductive material, such as copper, may be deposited in the openings in the patterned photoresist layer to form the conductive pillars **152**. The conductive material may be deposited using any suitable process, such as electroplating, sputtering, or electroless plating. The photoresist may be removed to expose the conductive pillars **152**. In another example, a photo-imagable dielectric may be used to form the conductive pillars **152**. In some embodiments, the insulating material **133** may be initially deposited on and over the top surfaces of the first die **114-1** and the conductive pillars **152**, then polished back to expose the top surface of the first die **114-1** and the conductive pillars **152**. The insulating material **133** may be formed using any suitable process, including lamination, or slit coating and curing. If the insulating material **133** is formed to completely cover the first die **114-1** and conductive pillars **152**, the insulating material **133** may be removed using any suitable technique, including grinding, or etching, such as a wet etch, a dry etch (e.g., a plasma etch), a wet blast, or a laser ablation (e.g., using excimer laser). In some embodiments, the thickness of the insulating material **133** may be minimized to reduce the etching time required. The die **114-1** and/or the die **114-2** may be functionally tested using the conductive contacts **132** to determine that the dies **114-1**, **114-2** are KGDs before further processing is performed.

(57) FIG. **8J** illustrates an assembly subsequent to inverting the assembly of FIG. **8I**. The assembly of FIG. **8J** may be a microelectronic assembly **100**, as shown, or further manufacturing operations may be performed on the microelectronic assembly **100** of FIG. **8J** to form other microelectronic assemblies **100**, for example, as shown in FIG. **3**. For example, the assembly of FIG. **8J** may be electrically coupled to a package substrate through DTP interconnects by printing a solder paste on the conductive contacts **132**, placing the assembly of FIG. **8J** on a package substrate using a pick-n-place tool, subjecting the solder paste to thermal reflow, and cleaning.

(58) FIGS. **9A-9D** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. **1**, in accordance with various embodiments. FIG. **9A** illustrates the assembly of FIG. **8E** subsequent to performing the processes as described above with reference to FIGS. **8A-8E**.

(59) FIG. **9B** illustrates an assembly subsequent to forming conductive pillars **152** on the second carrier **105-2**, depositing an insulating material **133** on and around the first die **114-1** and the

conductive pillars **152**, and forming an exposed HB interface **180** on a top surface of the insulating material **133**, the conductive pillars **152**, and the die **114-1** (e.g., on the active side metallization stack **122**), where the HB interface **180** includes HB contacts **110** surrounded by HB dielectric **108**. The conductive pillars **152** and the insulating material **133** may be formed as described above with reference to FIG. **8I**.

(60) FIG. **9C** illustrates an assembly subsequent to hybrid bonding a second die **114-2** and a third die **114-3** to the top surface of the assembly of FIG. **9B**, and depositing an insulating material **133** on and around the second and third dies **114-2**, **114-3**. In particular, HB interface **180** (not labeled) of the second die **114-2** and the third die **114-3** may be brought into contact with the HB interface of the first die **114-1** and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB regions **130-1** and **130-2**, respectively. The insulating material **133** may be deposited as described above with reference to FIG. **8I**. In some embodiments, the insulating material **133** on and around the second and third dies **114-2**, **114-3** may be omitted. In such an embodiment, the second and third dies **114-2**, **114-3** may be supported by the underlying structure (e.g., the assembly of FIG. **9B**). In some embodiments, a mechanical support substrate, such as a permanent carrier (not shown), may be attached to the top surface of the assembly of FIG. **9C** (e.g., the top surfaces of the second and third dies **114-2**, **114-3**) to provide further mechanical support.

(61) FIG. **9D** illustrates an assembly subsequent to removing the second carrier **105-2** and forming conductive contacts **132** for DTP interconnects on a bottom surface of the assembly of FIG. **9C**.

The dies **114-1**, **114-2**, **114-3** may be functionally tested using the conductive contacts **132** to determine that the dies **114-1**, **114-2**, **114-3** are KGDs before further processing is performed. The assembly of FIG. **9D** may be a microelectronic assembly **100**, as shown, or further manufacturing operations may be performed on the microelectronic assembly **100** of FIG. **9D** to form other microelectronic assemblies **100**, for example, as shown in FIG. **1**. For example, the assembly of FIG. **9D** may be electrically coupled to a package substrate through DTP interconnects by printing a solder paste on the conductive contacts **132**, placing the assembly of FIG. **9D** on a package substrate using a pick-n-place tool, subjecting the solder paste to thermal reflow, and cleaning.

(62) FIGS. **10A-10G** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. **5**, in accordance with various embodiments. FIG. **10A** illustrates an assembly of FIG. **8D** subsequent to performing the processes as described above with reference to FIGS. **8A-8D**, where the second carrier **105-2** mounted to the top surface includes a substrate **128** and TSVs **118** (e.g., the second carrier **105-2** becomes a permanent part of the microelectronic assembly **100** of FIG. **5**).

(63) FIG. **10B** illustrates an assembly subsequent to inverting the assembly of FIG. **10A**, removing the first carrier **105-1**, and forming an exposed HB interface **180** on a top surface of the die **114-1** (e.g., on the active side metallization stack **122**), where the HB interface **180** includes HB contacts **110** surrounded by HB dielectric **108**.

(64) FIG. **10C** illustrates an assembly subsequent to placing a second die **114-2** and a third die **114-3** on a third carrier **105-3** with a backside (e.g., non-active side) facing towards the third carrier **105-3**. The top surfaces of the second and third dies **114-2**, **114-3** may include an exposed HB interface **180-1**, **180-2**, respectively, where the HB interface **180** includes HB contacts **110** surrounded by HB dielectric **108**. In some embodiments, an insulating material **133** (not shown) may be deposited on and around the second and third dies **114-2**, **114-3**, as described above with reference to FIG. **8I**.

(65) FIG. **10D** illustrates an assembly subsequent to hybrid bonding the first die **114-1** (e.g., inverting the assembly of FIG. **10B**) to the second die **114-2** and the third die **114-3** (e.g., to the top surface of the assembly of FIG. **10C**). In particular, HB interface **180** (not labeled) of the first die **114-1** may be brought into contact with the HB interface of the second die **114-2** and the third die **114-3**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB regions **130-1** and **130-2**, respectively.

(66) FIG. **10E** illustrates an assembly subsequent to removing non-electrical material from the backside (e.g., the top surface) of the substrate **128** and revealing the top surface of the TSVs **118**. The non-electrical material may be removed using any suitable technique, including, for example, as described above with reference to FIG. **8B**.

(67) FIG. **10F** illustrates an assembly subsequent to forming conductive pillars **152** on the second and third dies **114-2**, **114-3**, depositing an insulating material **133** on and around the first die **114-1** and the conductive pillars **152**, and forming conductive contacts **132** for DTP interconnects on the top surface to the assembly. The dies **114-1**, **114-2**, **114-3** may be functionally tested using the conductive contacts **132** to determine that the dies **114-1**, **114-2**, **114-3** are KGDs before further processing is performed. In some embodiments, the insulating material **133** may be deposited on and around the second and third dies **114-2**, **114-3**.

(68) FIG. **10G** illustrates an assembly subsequent to inverting the assembly of FIG. **10F** and removing the third carrier **105-3**. The assembly of FIG. **10G** may be a microelectronic assembly **100**, as shown, or further manufacturing operations may be performed on the microelectronic assembly **100** of FIG. **10G** to form other microelectronic assemblies **100**, for example, as shown in FIG. **5**. For example, the assembly of FIG. **10G** may be electrically coupled to a package substrate through DTP interconnects by printing a solder paste on the conductive contacts **132**, placing the assembly of FIG. **10G** on a package substrate using a pick-n-place tool, subjecting the solder paste to thermal reflow, and cleaning.

(69) FIGS. **11A-11D** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. **7**, in accordance with various embodiments.

(70) FIG. **11A** is an assembly subsequent to performing the processes described above with reference to FIGS. **10A-10B** on first and second dies **114-1**, **114-2** and hybrid bonding the first and second dies **114-1**, **114-2** to a third die **114-3**. In particular, HB interface **180** (not labeled) of the first die **114-1** and the second die **114-2** may be brought into contact with the HB interface of the third die **114-3**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB regions **130-1** and **130-2**, respectively.

(71) FIG. **11B** illustrates an assembly subsequent to removing the non-electrical material from the top surface (e.g., backside) of the substrate **128** of the first and second dies **114-1**, **114-2** to reveal the top surfaces of the TSVs **118**, forming conductive pillars **152** on the top surface of the third die **114-3**, and depositing an insulating material **133** on and around the first die **114-1**, the second die **114-2**, and the conductive pillars **152**. In some embodiments, the non-electrical material from the top surface of the substrate **128** of the first and second dies **114-1**, **114-2** may be removed with the insulating material **133**. The conductive pillars **152** and the insulating material **133** may be formed using any suitable technique, including as described above with reference to FIG. **8I**. The insulating material **133** may be removed using any suitable technique, including as described above with reference to FIG. **8I**. The non-electrical material of the substrate **128** may be removed using any suitable technique, including as described above with reference to FIG. **8B**.

(72) FIG. **11C** illustrates an assembly subsequent to forming an RDL **148** on the top surface of the assembly of FIG. **11B**. The RDL **148** may include conductive contacts on a bottom surface coupled to the first and second dies **114-1**, **114-2** by the TSVs **118** in the substrate **128**, and conductive contacts **132** on the top surface for coupling to a package substrate by DTP interconnects. The RDL **148** may be manufactured using any suitable technique, such as a PCB technique, a redistribution layer technique, or damascene processing.

(73) FIG. **11D** illustrates an assembly subsequent to inverting the assembly of FIG. **11C**. The assembly of FIG. **11D** may itself be a microelectronic assembly **100**, as shown. Further manufacturing operations may be performed on the microelectronic assembly **100** of FIG. **11D** to form other microelectronic assemblies **100**, such as shown in FIG. **7**. For example, further processing may include depositing a solder resist layer, attaching solder balls, and electrically coupling a package substrate **102** to the bottom surface of the assembly of FIG. **11D** through DTP

interconnects **150**. The dies **114-1**, **114-2**, **114-3** may be functionally tested using the conductive contacts **132** to determine that the dies **114-1**, **114-2**, **114-3** are KGDs before further processing is performed.

(74) The microelectronic assemblies **100** disclosed herein may be used for any suitable application. For example, in some embodiments, a microelectronic assembly **100** may be used to enable very small form factor voltage regulation for field programmable gate array (FPGA) or processing units (e.g., a central processing unit, a graphics processing unit, a FPGA, a modem, an applications processor, etc.) especially in mobile devices and small form factor devices. In another example, the die **114** in a microelectronic assembly **100** may be a processing device (e.g., a central processing unit, a graphics processing unit, a FPGA, a modem, an applications processor, etc.).

(75) The microelectronic assemblies **100** disclosed herein may be included in any suitable electronic component. FIGS. **12-15** illustrate various examples of apparatuses that may include, or be included in, any of the microelectronic assemblies **100** disclosed herein.

(76) FIG. **12** is a top view of a wafer **1500** and dies **1502** that may be included in any of the microelectronic assemblies **100** disclosed herein (e.g., as any suitable ones of the dies **114**). The wafer **1500** may be composed of semiconductor material and may include one or more dies **1502** having IC structures formed on a surface of the wafer **1500**. Each of the dies **1502** may be a repeating unit of a semiconductor product that includes any suitable IC. After the fabrication of the semiconductor product is complete, the wafer **1500** may undergo a singulation process in which the dies **1502** are separated from one another to provide discrete “chips” of the semiconductor product. The die **1502** may be any of the dies **114** disclosed herein. The die **1502** may include one or more transistors (e.g., some of the transistors **1640** of FIG. **13**, discussed below), supporting circuitry to route electrical signals to the transistors, passive components (e.g., signal traces, resistors, capacitors, or inductors), and/or any other IC components. In some embodiments, the wafer **1500** or the die **1502** may include a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1502**. For example, a memory array formed by multiple memory devices may be formed on a same die **1502** as a processing device (e.g., the processing device **1802** of FIG. **15**) or other logic that is configured to store information in the memory devices or execute instructions stored in the memory array. In some embodiments, a die **1502** (e.g., a die **114**) may be a central processing unit, a radio frequency chip, a power converter, or a network processor. Various ones of the microelectronic assemblies **100** disclosed herein may be manufactured using a die-to-wafer assembly technique in which some dies **114** are attached to a wafer **1500** that include others of the dies **114**, and the wafer **1500** is subsequently singulated.

(77) FIG. **13** is a cross-sectional side view of an IC device **1600** that may be included in any of the microelectronic assemblies **100** disclosed herein (e.g., in any of the dies **114**). One or more of the IC devices **1600** may be included in one or more dies **1502** (FIG. **12**). The IC device **1600** may be formed on a die substrate **1602** (e.g., the wafer **1500** of FIG. **12**) and may be included in a die (e.g., the die **1502** of FIG. **12**). The die substrate **1602** may be a semiconductor substrate composed of semiconductor material systems including, for example, n-type or p-type materials systems (or a combination of both). The die substrate **1602** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some embodiments, the die substrate **1602** may be formed using alternative materials, which may or may not be combined with silicon, that include, but are not limited to, germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the die substrate **1602**. Although a few examples of materials from which the die substrate **1602** may be formed are described here, any material that may serve as a foundation for an IC device **1600** may be used. The die substrate

1602 may be part of a singulated die (e.g., the dies **1502** of FIG. **12**) or a wafer (e.g., the wafer **1500** of FIG. **12**).

(78) The IC device **1600** may include one or more device layers **1604** disposed on the die substrate **1602**. The device layer **1604** may include features of one or more transistors **1640** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the die substrate **1602**. The device layer **1604** may include, for example, one or more source and/or drain (S/D) regions **1620**, a gate **1622** to control current flow in the transistors **1640** between the S/D regions **1620**, and one or more S/D contacts **1624** to route electrical signals to/from the S/D regions **1620**. The transistors **1640** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1640** are not limited to the type and configuration depicted in FIG. **13** and may include a wide variety of other types and configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

(79) Each transistor **1640** may include a gate **1622** formed of at least two layers, a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric to improve its quality when a high-k material is used.

(80) The gate electrode may be formed on the gate dielectric and may include at least one p-type work function metal or n-type work function metal, depending on whether the transistor **1640** is to be a PMOS or a NMOS transistor. In some implementations, the gate electrode may consist of a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide), and any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

(81) In some embodiments, when viewed as a cross-section of the transistor **1640** along the source-channel-drain direction, the gate electrode may consist of a U-shaped structure that includes a bottom portion substantially parallel to the surface of the die substrate **1602** and two sidewall portions that are substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the die substrate **1602** and does not include sidewall portions substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, the gate electrode may consist of a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may consist of one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

(82) In some embodiments, a pair of sidewall spacers may be formed on opposing sides of the gate

stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition and etching process steps. In some embodiments, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.

(83) The S/D regions **1620** may be formed within the die substrate **1602** adjacent to the gate **1622** of each transistor **1640**. The S/D regions **1620** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the die substrate **1602** to form the S/D regions **1620**. An annealing process that activates the dopants and causes them to diffuse farther into the die substrate **1602** may follow the ion-implantation process. In the latter process, the die substrate **1602** may first be etched to form recesses at the locations of the S/D regions **1620**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **1620**. In some implementations, the S/D regions **1620** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the S/D regions **1620** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1620**.

(84) Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., transistors **1640**) of the device layer **1604** through one or more interconnect layers disposed on the device layer **1604** (illustrated in FIG. **13** as interconnect layers **1606-1610**). For example, electrically conductive features of the device layer **1604** (e.g., the gate **1622** and the S/D contacts **1624**) may be electrically coupled with the interconnect structures **1628** of the interconnect layers **1606-1610**. The one or more interconnect layers **1606-1610** may form a metallization stack (also referred to as an “ILD stack”) **1619** of the IC device **1600**.

(85) The interconnect structures **1628** may be arranged within the interconnect layers **1606-1610** to route electrical signals according to a wide variety of designs; in particular, the arrangement is not limited to the particular configuration of interconnect structures **1628** depicted in FIG. **13**. Although a particular number of interconnect layers **1606-1610** is depicted in FIG. **13**, embodiments of the present disclosure include IC devices having more or fewer interconnect layers than depicted.

(86) In some embodiments, the interconnect structures **1628** may include lines **1628a** and/or vias **1628b** filled with an electrically conductive material such as a metal. The lines **1628a** may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the die substrate **1602** upon which the device layer **1604** is formed. For example, the lines **1628a** may route electrical signals in a direction in and out of the page from the perspective of FIG. **13**. The vias **1628b** may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the die substrate **1602** upon which the device layer **1604** is formed. In some embodiments, the vias **1628b** may electrically couple lines **1628a** of different interconnect layers **1606-1610** together.

(87) The interconnect layers **1606-1610** may include a dielectric material **1626** disposed between the interconnect structures **1628**, as shown in FIG. **13**. In some embodiments, the dielectric material **1626** disposed between the interconnect structures **1628** in different ones of the interconnect layers **1606-1610** may have different compositions; in other embodiments, the composition of the dielectric material **1626** between different interconnect layers **1606-1610** may be the same.

(88) A first interconnect layer **1606** (referred to as Metal 1 or “M1”) may be formed directly on the

device layer **1604**. In some embodiments, the first interconnect layer **1606** may include lines **1628a** and/or vias **1628b**, as shown. The lines **1628a** of the first interconnect layer **1606** may be coupled with contacts (e.g., the S/D contacts **1624**) of the device layer **1604**.

(89) A second interconnect layer **1608** (referred to as Metal 2 or “M2”) may be formed directly on the first interconnect layer **1606**. In some embodiments, the second interconnect layer **1608** may include vias **1628b** to couple the lines **1628a** of the second interconnect layer **1608** with the lines **1628a** of the first interconnect layer **1606**. Although the lines **1628a** and the vias **1628b** are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **1608**) for the sake of clarity, the lines **1628a** and the vias **1628b** may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual damascene process) in some embodiments.

(90) A third interconnect layer **1610** (referred to as Metal 3 or “M3”) (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1608** according to similar techniques and configurations described in connection with the second interconnect layer **1608** or the first interconnect layer **1606**. In some embodiments, the interconnect layers that are “higher up” in the metallization stack **1619** in the IC device **1600** (i.e., farther away from the device layer **1604**) may be thicker.

(91) The IC device **1600** may include a solder resist material **1634** (e.g., polyimide or similar material) and one or more conductive contacts **1636** formed on the interconnect layers **1606-1610**. In FIG. **13**, the conductive contacts **1636** are illustrated as taking the form of bond pads. The conductive contacts **1636** may be electrically coupled with the interconnect structures **1628** and configured to route the electrical signals of the transistor(s) **1640** to other external devices. For example, solder bonds may be formed on the one or more conductive contacts **1636** to mechanically and/or electrically couple a chip including the IC device **1600** with another component (e.g., a circuit board). The IC device **1600** may include additional or alternate structures to route the electrical signals from the interconnect layers **1606-1610**; for example, the conductive contacts **1636** may include other analogous features (e.g., posts) that route the electrical signals to external components.

(92) In some embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include another metallization stack (not shown) on the opposite side of the device layer(s) **1604**. This metallization stack may include multiple interconnect layers as discussed above with reference to the interconnect layers **1606-1610**, to provide conductive pathways (e.g., including conductive lines and vias) between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**.

(93) In other embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include one or more TSVs through the die substrate **1602**; these TSVs may make contact with the device layer(s) **1604**, and may provide conductive pathways between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**.

(94) FIG. **14** is a cross-sectional side view of an IC device assembly **1700** that may include any of the microelectronic assemblies **100** disclosed herein. In some embodiments, the IC device assembly **1700** may be a microelectronic assembly **100**. The IC device assembly **1700** includes a number of components disposed on a circuit board **1702** (which may be, e.g., a motherboard). The IC device assembly **1700** includes components disposed on a first face **1740** of the circuit board **1702** and an opposing second face **1742** of the circuit board **1702**; generally, components may be disposed on one or both faces **1740** and **1742**. Any of the IC packages discussed below with reference to the IC device assembly **1700** may take the form of any suitable ones of the embodiments of the microelectronic assemblies **100** disclosed herein.

(95) In some embodiments, the circuit board **1702** may be a PCB including multiple metal layers

separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1702**. In other embodiments, the circuit board **1702** may be a non-PCB substrate. In some embodiments the circuit board **1702** may be, for example, a circuit board.

(96) The IC device assembly **1700** illustrated in FIG. **14** includes a package-on-interposer structure **1736** coupled to the first face **1740** of the circuit board **1702** by coupling components **1716**. The coupling components **1716** may electrically and mechanically couple the package-on-interposer structure **1736** to the circuit board **1702**, and may include solder balls (as shown in FIG. **14**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

(97) The package-on-interposer structure **1736** may include an IC package **1720** coupled to an interposer **1704** by coupling components **1718**. The coupling components **1718** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1716**. Although a single IC package **1720** is shown in FIG. **14**, multiple IC packages may be coupled to the interposer **1704**; indeed, additional interposers may be coupled to the interposer **1704**. The interposer **1704** may provide an intervening substrate used to bridge the circuit board **1702** and the IC package **1720**. The IC package **1720** may be or include, for example, a die (the die **1502** of FIG. **12**), an IC device (e.g., the IC device **1600** of FIG. **13**), or any other suitable component. Generally, the interposer **1704** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **1704** may couple the IC package **1720** (e.g., a die) to a set of ball grid array (BGA) conductive contacts of the coupling components **1716** for coupling to the circuit board **1702**. In the embodiment illustrated in FIG. **14**, the IC package **1720** and the circuit board **1702** are attached to opposing sides of the interposer **1704**; in other embodiments, the IC package **1720** and the circuit board **1702** may be attached to a same side of the interposer **1704**. In some embodiments, three or more components may be interconnected by way of the interposer **1704**.

(98) In some embodiments, the interposer **1704** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. In some embodiments, the interposer **1704** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some embodiments, the interposer **1704** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1704** may include metal interconnects **1708** and vias **1710**, including but not limited to TSVs **1706**. The interposer **1704** may further include embedded devices **1714**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1704**. The package-on-interposer structure **1736** may take the form of any of the package-on-interposer structures known in the art.

(99) The IC device assembly **1700** may include an IC package **1724** coupled to the first face **1740** of the circuit board **1702** by coupling components **1722**. The coupling components **1722** may take the form of any of the embodiments discussed above with reference to the coupling components **1716**, and the IC package **1724** may take the form of any of the embodiments discussed above with reference to the IC package **1720**.

(100) The IC device assembly **1700** illustrated in FIG. **14** includes a package-on-package structure **1734** coupled to the second face **1742** of the circuit board **1702** by coupling components **1728**. The package-on-package structure **1734** may include an IC package **1726** and an IC package **1732**

coupled together by coupling components **1730** such that the IC package **1726** is disposed between the circuit board **1702** and the IC package **1732**. The coupling components **1728** and **1730** may take the form of any of the embodiments of the coupling components **1716** discussed above, and the IC packages **1726** and **1732** may take the form of any of the embodiments of the IC package **1720** discussed above. The package-on-package structure **1734** may be configured in accordance with any of the package-on-package structures known in the art.

(101) FIG. **15** is a block diagram of an example electrical device **1800** that may include one or more of the microelectronic assemblies **100** disclosed herein. For example, any suitable ones of the components of the electrical device **1800** may include one or more of the IC device assemblies **1700**, IC devices **1600**, or dies **1502** disclosed herein, and may be arranged in any of the microelectronic assemblies **100** disclosed herein. A number of components are illustrated in FIG. **15** as included in the electrical device **1800**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the electrical device **1800** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

(102) Additionally, in various embodiments, the electrical device **1800** may not include one or more of the components illustrated in FIG. **15**, but the electrical device **1800** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1800** may not include a display device **1806**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **1806** may be coupled. In another set of examples, the electrical device **1800** may not include an audio input device **1824** or an audio output device **1808**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1824** or audio output device **1808** may be coupled.

(103) The electrical device **1800** may include a processing device **1802** (e.g., one or more processing devices). As used herein, the term “processing device” or “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **1802** may include one or more digital signal processors (DSPs), application-specific ICs (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The electrical device **1800** may include a memory **1804**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **1804** may include memory that shares a die with the processing device **1802**. This memory may be used as cache memory and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

(104) In some embodiments, the electrical device **1800** may include a communication chip **1812** (e.g., one or more communication chips). For example, the communication chip **1812** may be configured for managing wireless communications for the transfer of data to and from the electrical device **1800**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

(105) The communication chip **1812** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005

Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UM B) project (also referred to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1812** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1812** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1812** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1812** may operate in accordance with other wireless protocols in other embodiments. The electrical device **1800** may include an antenna **1822** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

(106) In some embodiments, the communication chip **1812** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1812** may include multiple communication chips. For instance, a first communication chip **1812** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1812** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **1812** may be dedicated to wireless communications, and a second communication chip **1812** may be dedicated to wired communications.

(107) The electrical device **1800** may include battery/power circuitry **1814**. The battery/power circuitry **1814** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1800** to an energy source separate from the electrical device **1800** (e.g., AC line power).

(108) The electrical device **1800** may include a display device **1806** (or corresponding interface circuitry, as discussed above). The display device **1806** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.

(109) The electrical device **1800** may include an audio output device **1808** (or corresponding interface circuitry, as discussed above). The audio output device **1808** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds.

(110) The electrical device **1800** may include an audio input device **1824** (or corresponding interface circuitry, as discussed above). The audio input device **1824** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

(111) The electrical device **1800** may include a GPS device **1818** (or corresponding interface circuitry, as discussed above). The GPS device **1818** may be in communication with a satellite-based system and may receive a location of the electrical device **1800**, as known in the art.

(112) The electrical device **1800** may include an other output device **1810** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1810** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

(113) The electrical device **1800** may include an other input device **1820** (or corresponding

interface circuitry, as discussed above). Examples of the other input device **1820** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

(114) The electrical device **1800** may have any desired form factor, such as a computing device or a hand-held, portable or mobile computing device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultra mobile personal computer, etc.), a desktop electrical device, a server, or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable computing device. In some embodiments, the electrical device **1800** may be any other electronic device that processes data.

(115) The following paragraphs provide various examples of the embodiments disclosed herein.

(116) Example 1 is a microelectronic assembly, including a first die, having a first surface and an opposing second surface, in a first layer, the first die including a first metallization stack at the first surface; a device layer, including a device, on the first metallization stack; a second metallization stack on the device layer; and an interconnect at the first surface electrically coupled to the first metallization stack; a conductive pillar in the first layer; and a second die, having a first surface and an opposing second surface, in a second layer on the first layer, wherein the first surface of the second die is coupled to the second surface of the first die by a hybrid bonding region and is coupled to the conductive pillar.

(117) Example 2 may include the subject matter of Example 1, and may further specify that the hybrid bonding region is a first hybrid bonding region, and may further include a third die, having a first surface and an opposing second surface, in the second layer, wherein the first surface of the third die is coupled to the second surface of the first die by a second hybrid bonding region.

(118) Example 3 may include the subject matter of Example 1 or 2, and may further specify that the interconnect is part of a power delivery network.

(119) Example 4 may include the subject matter of any of Examples 1-3, and may further include a substrate layer, including a micro-through silicon via (μ TSV), between the first metallization stack and the device layer.

(120) Example 5 may include the subject matter of Example 4, and may further specify that the μ TSV electrically couples the device in the device layer to the first metallization stack.

(121) Example 6 may include the subject matter of any of the Examples 1-5, and may further include a package substrate electrically coupled to the first surface of the first die by the interconnect and electrically coupled to the first surface of the second die by the conductive pillar.

(122) Example 7 may include the subject matter of any of Examples 1-6, and may further specify that the interconnect is a first interconnect, and may further include a third die, having a first surface with a second interconnect and an opposing second surface, electrically coupled to the first surface of the first die by the first interconnect and electrically coupled to the first surface of the second die by the conductive pillar.

(123) Example 8 may include the subject matter of Example 7, and may further specify that the second interconnect is part of a power delivery network.

(124) Example 9 may include the subject matter of Example 7, and may further include a package substrate electrically coupled to the first surface of the third die by the second interconnect.

(125) Example 10 may include the subject matter of any of Examples 1-9, and may further specify that conductive structures of the first metallization stack are thicker than conductive structures of the second metallization stack.

(126) Example 11 is a microelectronic assembly, including a first die, having a first surface and an opposing second surface, in a first layer, the first die including a substrate including a through-substrate via (TSV) at the first surface; a first metallization stack on the substrate; a device layer,

including a device, on the first metallization stack; a second metallization stack on the device layer; and an interconnect at the first surface electrically coupled to the first metallization stack by the TSV in the substrate; a conductive pillar in the first layer; and a second die, having a first surface and an opposing second surface, in a second layer on the first layer, wherein the first surface of the second die is coupled to the conductive pillar and to the second surface of the first die by a hybrid bonding region.

(127) Example 12 may include the subject matter of Example 11, and may further specify that the hybrid bonding region is a first hybrid bonding region, and may further include a third die, having a first surface and an opposing second surface, in the second layer, wherein the first surface of the third die is coupled to the second surface of the first die by a second hybrid bonding region.

(128) Example 13 may include the subject matter of Example 11 or 12, and may further specify that the interconnect is part of a power delivery network.

(129) Example 14 may include the subject matter of any of Examples 11-13, and may further specify that the substrate is a first substrate, and may further include a second substrate, including a micro-through silicon via (μ TSV), between the first metallization stack and the device layer.

(130) Example 15 may include the subject matter of Example 14, and may further specify that the μ TSV electrically couples the device in the device layer to the first metallization stack.

(131) Example 16 may include the subject matter of any of Examples 11-15, and may further include a package substrate electrically coupled to the first surface of the first die by the interconnect and electrically coupled to the first surface of the second die by the conductive pillar.

(132) Example 17 may include the subject matter of any of Examples 11-16, and may further specify that the interconnect is a first interconnect, and may further include a third die, having a first surface with a second interconnect and an opposing second surface, electrically coupled to the first surface of the first die by the first interconnect and electrically coupled to the first surface of the second die by the conductive pillar.

(133) Example 18 may include the subject matter of Example 17, and may further specify that the second interconnect is part of a power delivery network.

(134) Example 19 may include the subject matter of Example 17, and may further include a package substrate electrically coupled to the first surface of the third die by the second interconnect.

(135) Example 20 may include the subject matter of any of Examples 11-19, and may further specify that conductive structures of the first metallization stack are thicker than conductive structures of the second metallization stack.

(136) Example 21 is a microelectronic assembly, including a first die in a first dielectric layer, the first dielectric layer having a first surface and an opposing second surface, and the first die including a substrate including a through-substrate via (TSV) at the first surface; a first metallization stack on the substrate; a device layer, including a device, on the first metallization stack; a second metallization stack on the device layer; and first interconnects at the first surface electrically coupled to the first metallization stack by the TSV in the substrate; a second die in the first dielectric layer, the second die including a substrate including a through-substrate via (TSV) at the first surface; a first metallization stack on the substrate; a device layer, including a device, on the first metallization stack; a second metallization stack on the device layer; and second interconnects at the first surface electrically coupled to the first metallization stack by the TSV in the substrate; a conductive pillar in the first dielectric layer; a third die, in a second dielectric layer on the second surface of the first dielectric layer, electrically coupled to the conductive pillar, electrically coupled to the first die by a first hybrid bonding region at the second surface of the first dielectric layer, and electrically coupled to the second die by a second hybrid bonding region at the second surface of the first dielectric layer; and a redistribution layer (RDL), having a first surface and an opposing second surface, at the first surface of the first dielectric layer, wherein the second surface of the RDL is electrically coupled to the first surface of the first dielectric layer, and

wherein the first surface of the RDL includes third interconnects electrically coupled to the conductive pillar, the first interconnects, and the second interconnects by conductive pathways in the RDL.

(137) Example 22 may include the subject matter of Example 21, and may further specify that the first interconnects, the second interconnects, and the third interconnects are part of a power delivery network.

(138) Example 23 may include the subject matter of Example 21 or 22, and may further specify that the substrate of the first die is a first substrate, and may further include a second substrate, including a micro-through silicon via (μ TSV), between the first metallization stack and the device layer, wherein the μ TSV electrically couples the device in the device layer to the first metallization stack.

(139) Example 24 may include the subject matter of any of Examples 21-23, and may further specify that the substrate of the second die is a first substrate, and may further include a second substrate, including a μ TSV, between the first metallization stack and the device layer, wherein the μ TSV electrically couples the device in the device layer to the first metallization stack.

(140) Example 25 may include the subject matter of any of Examples 21-24, and may further include a package substrate electrically coupled to the first surface of the RDL by the third interconnects.

Claims

1. A microelectronic assembly, comprising: a first die, having a first surface and an opposing second surface, in a first layer, the first die including: a first metallization stack at the first surface; a device layer, including a device, on the first metallization stack; a second metallization stack on the device layer; and an interconnect at the first surface electrically coupled to the first metallization stack; a conductive pillar in the first layer; and a second die, having a first surface and an opposing second surface, in a second layer on the first layer, wherein the first surface of the second die is coupled to the second surface of the first die by a hybrid bonding region and is coupled to the conductive pillar.
2. The microelectronic assembly of claim 1, wherein the hybrid bonding region is a first hybrid bonding region, and further comprising: a third die, having a first surface and an opposing second surface, in the second layer, wherein the first surface of the third die is coupled to the second surface of the first die by a second hybrid bonding region.
3. The microelectronic assembly of claim 1, wherein the interconnect is part of a power delivery network.
4. The microelectronic assembly of claim 1, further comprising: a substrate layer, including a micro-through silicon via (μ TSV), between the first metallization stack and the device layer.
5. The microelectronic assembly of claim 4, wherein the μ TSV electrically couples the device in the device layer to the first metallization stack.
6. The microelectronic assembly of claim 1, further comprising: a package substrate electrically coupled to the first surface of the first die by the interconnect and electrically coupled to the first surface of the second die by the conductive pillar.
7. The microelectronic assembly of claim 1, wherein conductive structures of the first metallization stack are thicker than conductive structures of the second metallization stack.
8. A microelectronic assembly, comprising: a first die, having a first surface and an opposing second surface, in a first layer, the first die including: a substrate including a through-substrate via (TSV) at the first surface; a first metallization stack on the substrate; a device layer, including a device, on the first metallization stack; a second metallization stack on the device layer; and an interconnect at the first surface electrically coupled to the first metallization stack by the TSV in the substrate; a conductive pillar in the first layer; and a second die, having a first surface and an

opposing second surface, in a second layer on the first layer, wherein the first surface of the second die is coupled to the conductive pillar and to the second surface of the first die by a hybrid bonding region.

9. The microelectronic assembly of claim 8, wherein the interconnect is part of a power delivery network.

10. The microelectronic assembly of claim 8, wherein the substrate is a first substrate, and further comprising: a second substrate, including a micro-through silicon via (μ TSV), between the first metallization stack and the device layer.

11. The microelectronic assembly of claim 10, wherein the μ TSV electrically couples the device in the device layer to the first metallization stack.

12. The microelectronic assembly of claim 8, further comprising: a package substrate electrically coupled to the first surface of the first die by the interconnect and electrically coupled to the first surface of the second die by the conductive pillar.

13. The microelectronic assembly of claim 8, further comprising: a third die, having a first surface and an opposing second surface, wherein the second surface of the third die is electrically coupled to the first surface of the first die by the interconnect and electrically coupled to the first surface of the second die by the conductive pillar.

14. The microelectronic assembly of claim 13, wherein the first surface of the third die further includes a second interconnect, and the microelectronic assembly further comprises: a package substrate electrically coupled to the first surface of the third die by the second interconnect.

15. The microelectronic assembly of claim 14, wherein the second interconnect is part of a power delivery network.

16. The microelectronic assembly of claim 8, wherein conductive structures of the first metallization stack are thicker than conductive structures of the second metallization stack.

17. A microelectronic assembly, comprising: a first die in a first dielectric layer, the first dielectric layer having a first surface and an opposing second surface, and the first die including: a substrate including a through-substrate via (TSV) at the first surface; a first metallization stack on the substrate; a device layer, including a device, on the first metallization stack; a second metallization stack on the device layer; and first interconnects at the first surface electrically coupled to the first metallization stack by the TSV in the substrate; a second die in the first dielectric layer, the second die including: a substrate including a through-substrate via (TSV) at the first surface; a first metallization stack on the substrate; a device layer, including a device, on the first metallization stack; a second metallization stack on the device layer; and second interconnects at the first surface electrically coupled to the first metallization stack by the TSV in the substrate; a conductive pillar in the first dielectric layer; a third die, in a second dielectric layer on the second surface of the first dielectric layer, electrically coupled to the conductive pillar, electrically coupled to the first die by a first hybrid bonding region at the second surface of the first dielectric layer, and electrically coupled to the second die by a second hybrid bonding region at the second surface of the first dielectric layer; and a redistribution layer (RDL), having a first surface and an opposing second surface, at the first surface of the first dielectric layer, wherein the second surface of the RDL is electrically coupled to the first surface of the first dielectric layer, and wherein the first surface of the RDL includes third interconnects electrically coupled to the conductive pillar, the first interconnects, and the second interconnects by conductive pathways in the RDL.

18. The microelectronic assembly of claim 17, wherein the first interconnects, the second interconnects, and the third interconnects are part of a power delivery network.

19. The microelectronic assembly of claim 17, wherein the substrate of the first die is a first substrate, and further comprising: a second substrate, including a micro-through silicon via (μ TSV), between the first metallization stack and the device layer, wherein the μ TSV electrically couples the device in the device layer to the first metallization stack.

20. The microelectronic assembly of claim 17, further comprising: a package substrate electrically coupled to the first surface of the RDL by the third interconnects.
